Probability & Statistics

Revision Sheet

Every topic on its own page • formulas, method, worked example, traps

How to use this sheet

- **One topic per page.** Nothing splits across pages — read a page, close the book, and try to reproduce the formula box from memory.
- **Read in this order on each page:** the italic line (what the topic *is*) → the formula box → "how to approach" → the worked example → the traps.
- **The dashed "My notes" box is yours.** Add the variant that caught you out, a formula your coaching notes state differently, or the page number of a question you got wrong.
- **Traps in the orange box are the marks people actually lose.** Re-read only those the night before the exam.

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1. Counting & Combinatorics

Before any probability question you must be able to count the outcomes. Decide one thing first: does order matter?

KEY FORMULAS

Order matters / order doesn't

$$^nP_r = \frac{n!}{(n-r)!} \qquad ^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

Arrangements with repeated items

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}$$

Inclusion–exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = \sum |A| - \sum |A \cap B| + |A \cap B \cap C|$$

WHAT YOU MUST KNOW

- **Onto functions** n-set $\to$ m-set: $\sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k)^n$ (pure formula recall).
- **Derangements** (nothing in its own place): $D_n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$; D_1=0, D_2=1, D_3=2, D_4=9.
- **No two consecutive 1s** in a length-n binary string: count $= F_{n+2}$ — set up $a_n = a_{n-1} + a_{n-2}$, never list.

HOW TO APPROACH A QUESTION

1. Ask **"does order matter?"** $\to$ permutation if yes, combination if no.
2. Ask **"are repeats allowed?"** $\to$ changes the formula completely.
3. *At least* / *none* / *not* $\Rightarrow$ try the **complement** first.
4. If the answer depends on the previous case, set up a **recurrence** (Fibonacci-type) instead of enumerating.

WORKED EXAMPLE

Q. How many 4-digit numbers with distinct digits from $\{1, 3, 4, 6, 7\}$ are divisible by 3?

Digit sum must be divisible by 3. All five sum to 21, so dropping digit d leaves $21 - d$ — divisible by 3 only for $d = 3$ or $d = 6$.

$$2 \text{ choices of dropped digit} \times 4! \text{ arrangements} = 2 \times 24 = \mathbf{48}$$

WATCH OUT

- Mixing P and C: "how many ways to *choose* a committee" is C; "how many ways to *rank*" is P.
- Forgetting that identical objects need the $\frac{n!}{n_1!n_2!\cdots}$ correction.
- Trying to list cases for "no two consecutive" problems — always a recurrence.

MY NOTES

2. Probability Axioms & Basic Rules

The rules are short. Almost every mistake here is confusing 'mutually exclusive' with 'independent' — they are opposites in spirit.

KEY FORMULAS

Axioms

$$0 \le P(A) \le 1, \qquad P(\Omega) = 1, \qquad P(\varnothing) = 0$$

Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Complement

$$P(A^c) = 1 - P(A)$$

Mutually exclusive (disjoint)

$$P(A \cap B) = 0 \;\Rightarrow\; P(A \cup B) = P(A) + P(B)$$

Independent

$$P(A \cap B) = P(A)\,P(B)$$

WHAT YOU MUST KNOW

- **Mutually exclusive ≠ independent.** If A, B are disjoint and both have $P > 0$, then knowing A happened tells you B did *not* — so they are strongly *dependent*.
- For equally likely outcomes, $P(A) = \dfrac{\text{favourable}}{\text{total}}$ — this is where your counting work pays off.
- Union bound: $P\left(\bigcup_i A_i\right) \le \sum_i P(A_i)$, always true.
- Pairwise independence does *not* imply mutual independence of three or more events.

HOW TO APPROACH A QUESTION

1. Write down the sample space size first (or realise you don't need it).
2. If the words are "*or*" → addition rule; "*and*" → multiplication (check independence first).
3. If "at least one" appears, compute $1 - P(\text{none})$.
4. Only multiply $P(A)P(B)$ after you have *justified* independence — do not assume it.

WORKED EXAMPLE

Q. A fair die is rolled twice. A = "first roll is even", B = "sum is 7". Independent?
$P(A) = \frac{1}{2}$, $P(B) = \frac{6}{36} = \frac{1}{6}$. Pairs with an even first roll summing to 7: $(2,5), (4,3), (6,1)$, so $P(A \cap B) = \frac{3}{36} = \frac{1}{12}$.

$$P(A)P(B) = \tfrac{1}{2} \cdot \tfrac{1}{6} = \tfrac{1}{12} = P(A \cap B) \;\Rightarrow\; \textbf{independent}$$

WATCH OUT

- Assuming disjoint events are independent (they are not).
- Adding probabilities of events that can happen together, forgetting $-P(A \cap B)$.

MY NOTES

3. Conditional Probability & Bayes

Your single most-tested weak area. The whole game is: a rare event stays rare even after a positive test. Count people, don't juggle symbols.

KEY FORMULAS

Definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

Multiplication rule

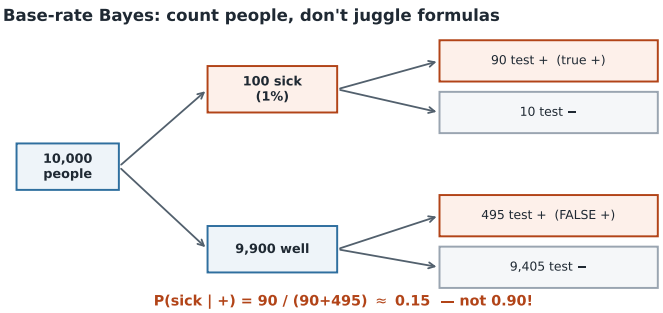
$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

Total probability (partition B_i)

$$P(A) = \sum_i P(A \mid B_i)\, P(B_i)$$

Bayes' theorem

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}$$



HOW TO APPROACH A QUESTION

1. **Imagine 10,000 people** (or 1,000). Turn every percentage into a count — the tree above.
2. Fill the four boxes: true positives, false negatives, false positives, true negatives.
3. The answer is $\dfrac{\text{the box you want}}{\text{the row/column total that matches the condition}}$.

WORKED EXAMPLE

Q. 1% have a disease. Test is 90% sensitive, 5% false-positive rate. Given a positive test, $P(\text{sick})$?

$$P = \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.05 \times 0.99} = \frac{0.009}{0.0585} \approx \mathbf{0.15}$$

Only about 15% — because the 99% healthy group produces far more false positives than the 1% sick group produces true positives.

WATCH OUT

- **Base-rate neglect**: answering 0.90 because the test is "90% accurate". The prior dominates.
- Confusing $P(A \mid B)$ with $P(B \mid A)$ — they are wildly different numbers.
- Forgetting to divide by the *total* probability of the evidence (the denominator).

MY NOTES

4. Random Variables: PMF, PDF, CDF

A random variable turns outcomes into numbers. Everything after this depends on knowing which of the three functions you have been handed.

KEY FORMULAS

Discrete

$$p(x) = P(X = x), \qquad \sum_x p(x) = 1$$

Continuous

$$f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

Probability of an interval

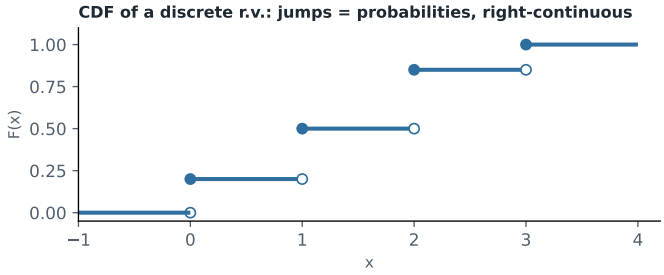
$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

CDF and its derivative

$$F(x) = P(X \leq x), \qquad F'(x) = f(x)$$

WHAT YOU MUST KNOW

- F is **non-decreasing**, right-continuous, with $F(-\infty) = 0$ and $F(+\infty) = 1$.
- For a **discrete** X, F is a step function; the *jump* at x equals $P(X = x)$.
- For a **continuous** X, $P(X = a) = 0$ for any single point, so $\leq$ and $<$ give the same answer.



HOW TO APPROACH A QUESTION

1. Identify discrete vs continuous — sums vs integrals.
2. If given f with an unknown constant, set $\int f = 1$ and solve for it first.
3. Given F, differentiate for f; given f, integrate from the left end of the support.

WORKED EXAMPLE

Q. $f(x) = cx$ on $[0, 2]$, zero elsewhere. Find c and $P(X \leq 1)$.

$$\int_0^2 cx\,dx = c \cdot \frac{4}{2} = 2c = 1 \;\Rightarrow\; c = \tfrac{1}{2}$$

$$P(X \leq 1) = \int_0^1 \tfrac{x}{2}\,dx = \tfrac{1}{4} = \mathbf{0.25}$$

WATCH OUT

- Treating $f(x)$ as a probability — a density can exceed 1; only its integral is a probability.
- Forgetting the support: integrating over $(-\infty, \infty)$ when f is only non-zero on part of it.

MY NOTES

5. Expectation, Variance & Moments

Expectation is linear — always. Variance is not: it picks up a covariance term unless the variables are independent.

KEY FORMULAS

Definition

$$E[X] = \sum_x x\,p(x) \qquad \text{or} \qquad E[X] = \int x f(x)\,dx$$

LOTUS (no need to find the law of $g(X)$)

$$E[g(X)] = \sum_x g(x)p(x)$$

Variance — the workhorse form

$$\mathrm{Var}(X) = E[X^2] - \big(E[X]\big)^2$$

Linear transforms

$$E[aX+b] = aE[X]+b, \qquad \mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$$

Sum of two variables

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$$

Law of total expectation

$$E[X] = E\big[E[X \mid Y]\big]$$

WHAT YOU MUST KNOW

- $E[\,\cdot\,]$ is **always** linear: $E[aX+bY] = aE[X]+bE[Y]$, *even if X, Y are dependent.*
- Independence gives $E[XY] = E[X]E[Y]$ and kills the covariance term.
- Markov: $P(X \geq a) \leq E[X]/a$. Chebyshev: $P(|X-\mu| \geq k\sigma) \leq 1/k^2$.

HOW TO APPROACH A QUESTION

1. For Var, compute $E[X^2]$ and $E[X]$ separately — almost never expand $(X-\mu)^2$.
2. For "expected number of ...", write $X = \sum_i I_i$ as a sum of **indicators** and use linearity. This works even when the I_i are dependent.
3. For a two-stage experiment ("first pick a box, then draw"), use the law of total expectation.

WORKED EXAMPLE

Q. 5 letters go into 5 envelopes at random. Expected number in the correct envelope?
Let $I_i = 1$ if letter i is correct. $E[I_i] = P(\text{letter } i \text{ correct}) = \frac{1}{5}$.

$$E\Big[\sum_{i=1}^{5} I_i\Big] = \sum_{i=1}^{5} E[I_i] = 5 \cdot \tfrac{1}{5} = \mathbf{1}$$

The I_i are dependent — linearity does not care.

WATCH OUT

- Writing $\mathrm{Var}(X+Y) = \mathrm{Var}X + \mathrm{Var}Y$ without checking independence.
- Using $\mathrm{Var}(aX) = a\,\mathrm{Var}(X)$; it is a^2.
- Assuming $E[1/X] = 1/E[X]$ or $E[X^2] = (E[X])^2$ — both false.

MY NOTES

6. Joint Distributions, Covariance & Correlation

Marked as your biggest gap: joint densities on a triangular region. The formula is easy; the integration limits are the entire question.

KEY FORMULAS

Covariance

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Correlation (always in $[-1,1]$)

$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\,\sigma_Y}$$

Marginal from a joint density

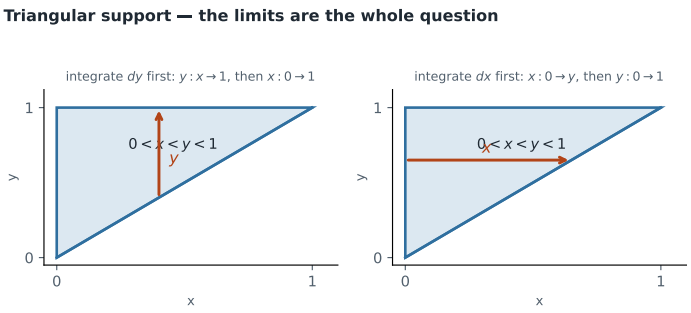
$$f_X(x) = \int f(x,y)\,dy$$

Conditional density

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}$$

WHAT YOU MUST KNOW

- Independent $\Rightarrow$ Cov $= 0$. **The converse is FALSE**: e.g. $X \sim U(-1,1)$ and $Y = X^2$ have Cov $= 0$ but are clearly dependent.
- A **triangular support means dependence** — the range of y depends on x, so they can never be independent.



HOW TO APPROACH A QUESTION

1. **Draw the region first.** Always. Shade $0 < x < y < 1$ before writing a single integral.
2. Choose the integration order that gives the simpler limits (see the two pictures above).
3. For a marginal $f_X(x)$: fix x, sweep y across the region — here y runs from x to 1.

WORKED EXAMPLE

Q. $f(x,y) = 2$ on $0 < x < y < 1$. Find $f_X(x)$ and $E[X]$.

$$f_X(x) = \int_{y=x}^{1} 2\,dy = 2(1-x), \qquad 0 < x < 1$$

$$E[X] = \int_0^1 x \cdot 2(1-x)\,dx = 2\left(\tfrac{1}{2} - \tfrac{1}{3}\right) = \mathbf{\tfrac{1}{3}}$$

WATCH OUT

- Using limits 0 to 1 for *both* variables — that is a square, not the triangle.
- Concluding independence from Cov $= 0$.

MY NOTES

7. Standard Discrete Distributions

Four distributions cover nearly every discrete question. Recognise the story, then the mean and variance are free.

KEY FORMULAS

Bernoulli(p) — one trial

$$E = p, \qquad \text{Var} = p(1-p)$$

Binomial(n, p) — successes in n fixed trials

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E = np, \qquad \text{Var} = np(1-p)$$

Geometric(p) — trials until the first success

$$P(X = k) = (1-p)^{k-1} p, \quad E = \frac{1}{p}, \quad \text{Var} = \frac{1-p}{p^2}$$

Poisson(λ) — rare events in a window

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad E = \text{Var} = \lambda$$

WHAT YOU MUST KNOW

- **Poisson window rescaling**: rate λ per unit time over a window of length t gives parameter λt. Half the Poisson questions test only this.
- **Recovering n and p**: from $\mu = np$ and $\sigma^2 = np(1-p)$, divide to get $1 - p = \sigma^2/\mu$, so $p = 1 - \sigma^2/\mu$ and $n = \mu/p$.
- Geometric is **memoryless**: $P(X > m + n \mid X > m) = P(X > n)$.
- "Exactly on the k-th trial" $\Rightarrow$ geometric; "exactly k out of n" $\Rightarrow$ binomial.

HOW TO APPROACH A QUESTION

1. Read the story: fixed number of trials? $\rightarrow$ binomial. Waiting for the first success? $\rightarrow$ geometric. Counting events in an interval? $\rightarrow$ Poisson.
2. Write down n, p or λ explicitly before computing anything.
3. For Poisson, rescale λ to the window the question actually asks about.
4. "At least one" $= 1 - P(0)$ — with Poisson that is $1 - e^{-\lambda}$.

WORKED EXAMPLE

Q. Calls arrive at 3 per hour (Poisson). P(at least one call in 20 minutes)?

20 minutes is $\frac{1}{3}$ hour, so $\lambda = 3 \times \frac{1}{3} = 1$.

$$P(X \geq 1) = 1 - P(0) = 1 - e^{-1} \approx \mathbf{0.632}$$

WATCH OUT

- Using the hourly rate for a sub-hour window without rescaling.
- Confusing "k-th trial is the first success" (geometric) with "k successes" (binomial).
- For binomial variance, writing np instead of $np(1-p)$.

MY NOTES

8. Standard Continuous Distributions

Three shapes: flat (uniform), decaying (exponential), bell (normal). Memorylessness and standardisation are the two ideas actually examined.

KEY FORMULAS

Uniform(a, b)

$$f = \frac{1}{b-a}, \qquad E = \frac{a+b}{2}, \qquad \text{Var} = \frac{(b-a)^2}{12}$$

Exponential(λ)

$$f = \lambda e^{-\lambda x}, \qquad E = \frac{1}{\lambda}, \qquad \text{Var} = \frac{1}{\lambda^2}$$

Normal(μ, σ^2)

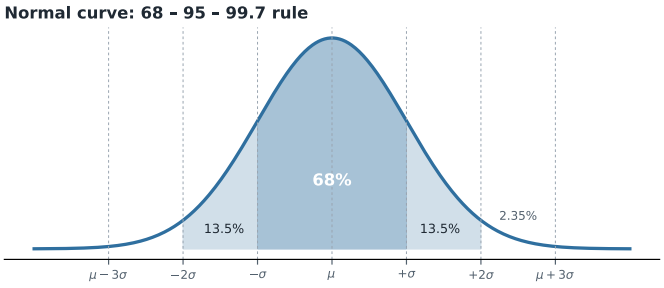
$$f = \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$$

Standardise before using tables

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1)$$

WHAT YOU MUST KNOW

- Exponential is **memoryless**: $P(X > s+t \mid X > s) = P(X > t)$; and $P(X > x) = e^{-\lambda x}$.
- Normal is symmetric about μ: $P(Z < -a) = P(Z > a) = 1 - \Phi(a)$.



HOW TO APPROACH A QUESTION

1. **Standardise first** — convert to a Z statement, then use symmetry.
2. For exponential, convert to the survival form $P(X > x) = e^{-\lambda x}$ immediately.
3. For uniform, probability is just the ratio of lengths.

WORKED EXAMPLE

Q. $X \sim N(100,\, 15^2)$. Find $P(85 < X < 130)$.

$$z_1 = \frac{85-100}{15} = -1, \qquad z_2 = \frac{130-100}{15} = 2$$

$$P(-1 < Z < 2) = 0.475 + 0.34 = \mathbf{0.815} \quad (\approx 81.5\%)$$

WATCH OUT

- Adding standard deviations instead of variances when summing normals.
- Forgetting that exponential's parameter is a *rate*: mean $= 1/\lambda$, not λ.

MY NOTES

9. Descriptive Statistics

Easy marks. The only real decisions are sample-vs-population divisor and how outliers move the mean but not the median.

KEY FORMULAS

Mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Population vs sample variance

$$\sigma^2 = \frac{1}{n}\sum(x_i - \mu)^2 \qquad s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$$

Standardisation (z-score)

$$z = \frac{x - \mu}{\sigma}$$

Min–max normalisation

$$x' = \frac{x - \min}{\max - \min} \in [0, 1]$$

WHAT YOU MUST KNOW

- **Median** = middle value once sorted (average the two middles when n is even). **Mode** = most frequent.
- The **mean is dragged by outliers**; the median is not. A question with one huge value is testing exactly this.
- IQR $= Q_3 - Q_1$; outliers lie beyond $Q_1 - 1.5\,\text{IQR}$ or $Q_3 + 1.5\,\text{IQR}$.
- After z-scoring, mean $= 0$ and sd $= 1$. After min–max, range $= [0, 1]$.

HOW TO APPROACH A QUESTION

1. **Sort the data first** — median, quartiles and mode all need it.
2. Decide n or $n - 1$: "sample" in the wording $\Rightarrow n - 1$.
3. For a shifted/scaled dataset, use $\text{Var}(aX + b) = a^2\text{Var}(X)$ instead of recomputing.

WORKED EXAMPLE

Q. Data: $9, 18, 11, 14, 15, 17, 10, 69, 11, 13$. Find the median and comment on the mean.
Sorted: $9, 10, 11, 11, 13, 14, 15, 17, 18, 69$. With $n = 10$, median $= \frac{13+14}{2} = \mathbf{13.5}$.
The mean is 18.7 — pulled far above every value except 69. **The median is the honest summary here.**

WATCH OUT

- Using n when the question says "sample standard deviation" (needs $n - 1$).
- Forgetting to sort before taking the median.

MY NOTES

10. CLT, Estimation & Confidence Intervals

The CLT is why the normal distribution is everywhere: averages become normal even when the data are not.

KEY FORMULAS

Central limit theorem

$$\bar{X} \dot{\sim} N\left(\mu,\ \frac{\sigma^2}{n}\right) \quad \text{for large } n$$

Standard error of the mean

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

CI when σ is known

$$\bar{x} \ \pm \ z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$$

CI when σ is unknown

$$\bar{x} \ \pm \ t_{\alpha/2,\,n-1}\,\frac{s}{\sqrt{n}}$$

WHAT YOU MUST KNOW

- Critical values worth memorising: $90\% \to 1.645$, $95\% \to 1.96$, $99\% \to 2.576$.
- The interval **narrows like** $1/\sqrt{n}$: to halve the width you need *four times* the data.
- An estimator is unbiased if $E[\hat{\theta}] = \theta$; $\text{MSE} = \text{bias}^2 + \text{variance}$.
- $\bar{X}$ is always unbiased for μ; s^2 (with $n-1$) is unbiased for σ^2.

HOW TO APPROACH A QUESTION

1. Identify σ known (use z) or unknown (use t with $n-1$ degrees of freedom).
2. Write $\text{SE} = \sigma/\sqrt{n}$ before anything else — most errors are a forgotten $\sqrt{n}$.
3. Multiply SE by the critical value to get the margin of error, then $\pm$ around $\bar{x}$.

WORKED EXAMPLE

Q. $n = 64$, $\bar{x} = 50$, $\sigma = 8$. Give a 95% confidence interval for μ.

$$\text{SE} = \frac{8}{\sqrt{64}} = 1, \qquad \text{margin} = 1.96 \times 1 = 1.96$$

$$\text{CI} = 50 \pm 1.96 = (\mathbf{48.04},\ \mathbf{51.96})$$

WATCH OUT

- Using σ instead of $\sigma/\sqrt{n}$ — the CI is about the *mean*, not one observation.
- Saying "95% probability that μ lies here". μ is fixed; the *interval* is random.

MY NOTES

11. Hypothesis Testing

You assume nothing is happening (H0), then ask how surprising the data would be. Small p-value means surprising, so reject.

KEY FORMULAS

z-test for a mean (σ known)

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

t-test for a mean (σ unknown)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \text{df} = n - 1$$

Chi-squared goodness of fit

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

WHAT YOU MUST KNOW

- **Type I error** (α): rejecting a *true* H_0 — a false alarm.
- **Type II error** (β): failing to reject a *false* H_0 — a miss. Power $= 1 - \beta$.
- Decision rule: $p < \alpha \Rightarrow$ **reject** H_0.
- Two-tailed tests split α across both tails ($\alpha/2$ each); one-tailed puts it all in one.

HOW TO APPROACH A QUESTION

1. State H_0 and H_1 in symbols before touching numbers.
2. Pick the test: mean with known $\sigma \to z$; mean with unknown $\sigma \to t$; counts/categories $\to \chi^2$.
3. Compute the statistic, compare with the critical value (or compare p with α).
4. State the conclusion **in context**, not just "reject H_0".

WORKED EXAMPLE

Q. $\mu_0 = 100$, $\bar{x} = 104$, $\sigma = 10$, $n = 25$. Test at $\alpha = 0.05$ (two-tailed).

$$z = \frac{104 - 100}{10/\sqrt{25}} = \frac{4}{2} = 2.0$$

Critical value for a two-tailed 5% test is 1.96. Since $2.0 > 1.96$, **reject** H_0 — the mean differs significantly from 100 (just barely).

WATCH OUT

- Using σ rather than $\sigma/\sqrt{n}$ in the denominator.
- Comparing a two-tailed statistic against a one-tailed critical value.
- Concluding "H_0 is true" — you only ever *fail to reject* it.

MY NOTES